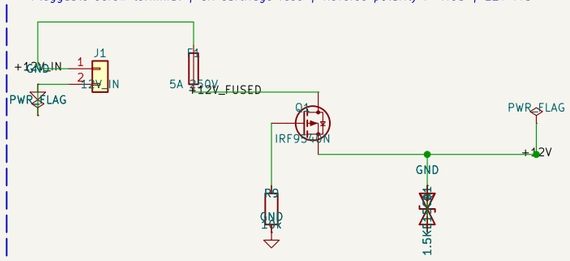
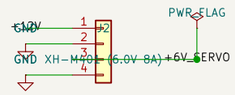
1. POWER INPUT & 12V PROTECTION

Pluggable screw terminal | 5A cartridge fuse | Reverse polarity P-MOS | 12V TVS

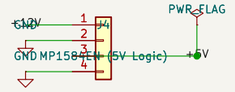


2. STEP-DOWN BUCK CONVERTERS (BREAKOUT MODULES)

J2: Servo Buck XH-M401 (XL4016E1 8A, Preset 6.0V) | J4: 5V Logic (MP1584EN) | J3: Aux Buck (DNP)

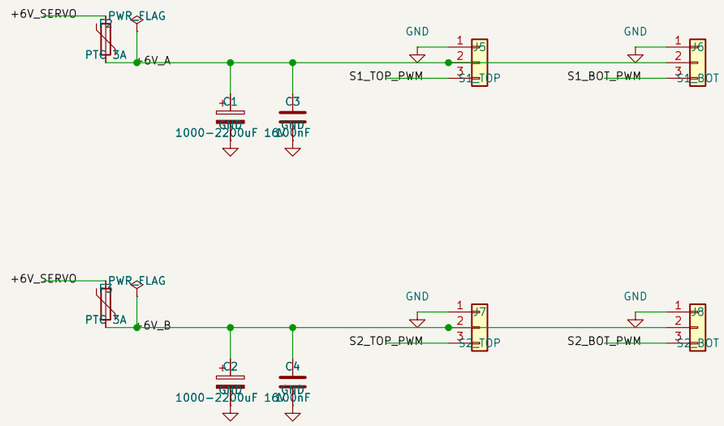






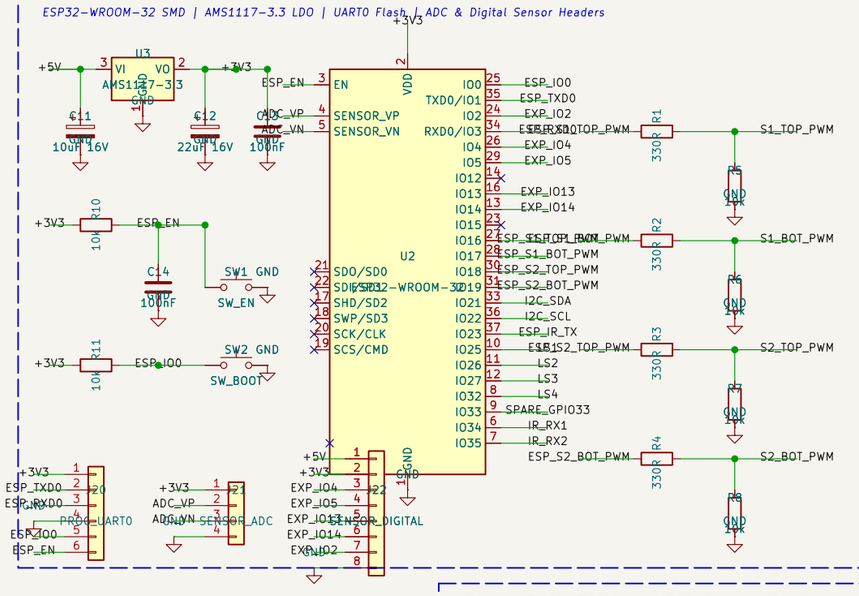
3. SERVO POWER PROTECTION & CONNECTORS

Rail A (Col 1) & Rail B (Col 2): PTC fuse | 1000-2200uF bulk cap | 100nF | MG996R 360 Headers



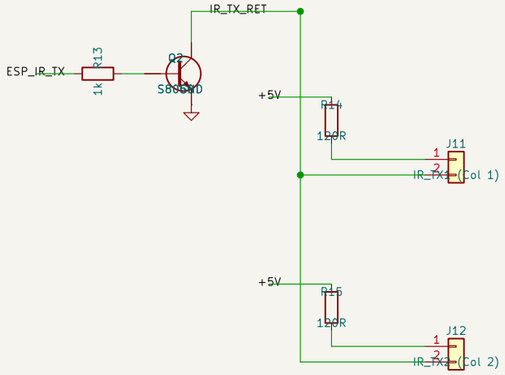
4. ESP32-WROOM-32 SMD MCU, 3.3V LDO & EXPANSION

ESP32-WROOM-32 SMD | AMS1117-3.3 LDO | UART0 Flash / ADC & Digital Sensor Headers



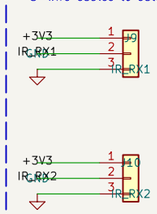
5. IR TRANSMITTER DRIVER CIRCUIT

GPIO23 -> NPN Driver (S8050/2N2222) | 2x JST-XH 2-pin connectors (1 per column)



6. IR RECEIVER HEADERS

3-wire cables to satellite RX sensor boards (38kHz TSOP/VS1838B)



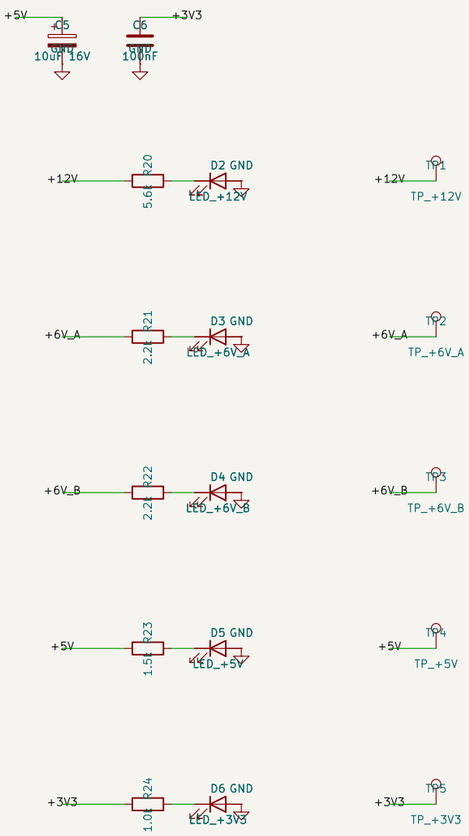
7. LIMIT SWITCHES (OPTIONAL - DNP)

4x JST-XH 2-pin | 10k pull-up to 3V3 | 100nF filter (Not populated in Rev A)



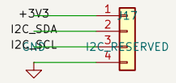
8. DIAGNOSTICS & RAIL DECOUPLING

Power LEDs and dedicated test points for all rails (+12V, +6V_A, +6V_B, +5V, +3V3, GND)



9. AUXILIARY / I2C EXPANSION HEADER

Reserved 4-pin header for optional external display, sensors or I2C expander



2-layer THT | Dual-column FIFO servo & IR control
Code-generated native KiCad 10 schematic
FV Project

Sheet: /
File: FV-CTRL_RevA.kicad_sch

Title: FV-CTRL Rev A (Fruit Vending Controller)

Size: A3	Date: 2026-10-16	**Rev: Rev A (Engineering Draft**
KiCad E.D.A. 10.0.7		Id: 1/1